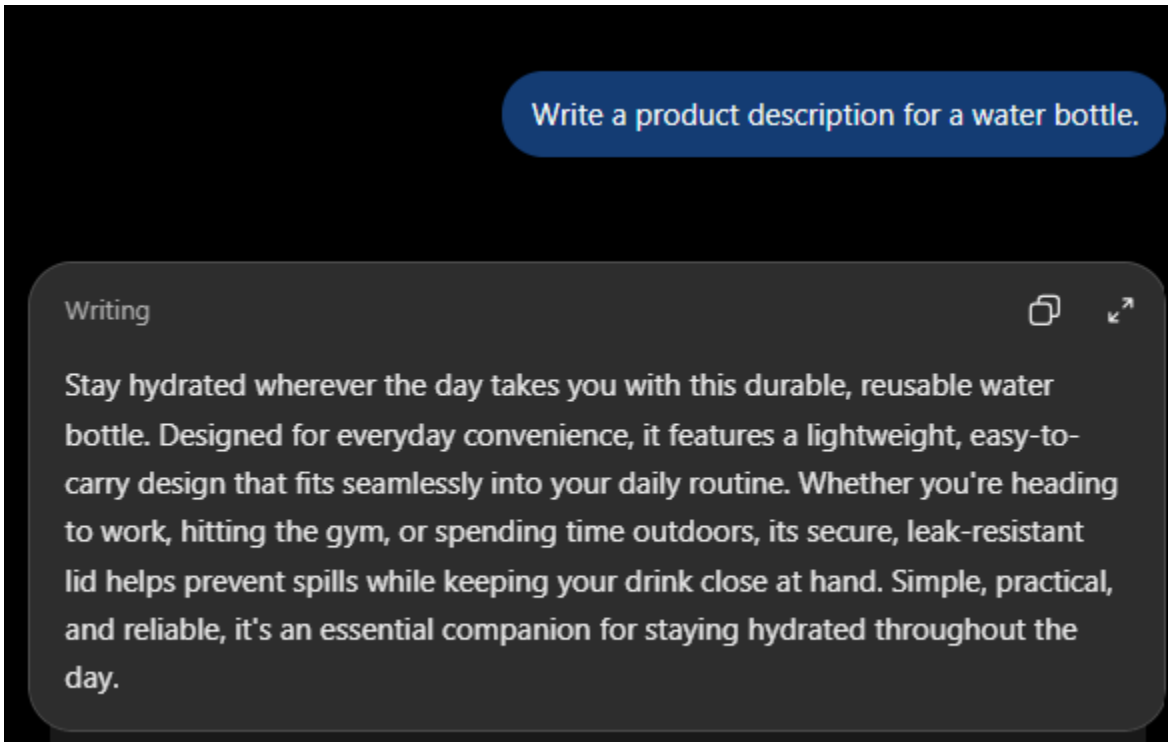
Kevin Hernandez
2026 November 2

Prompt Engineering Assignment 1

Welcome to my write-up, where we will be experimenting with prompt engineering and exploring techniques to produce better outputs.

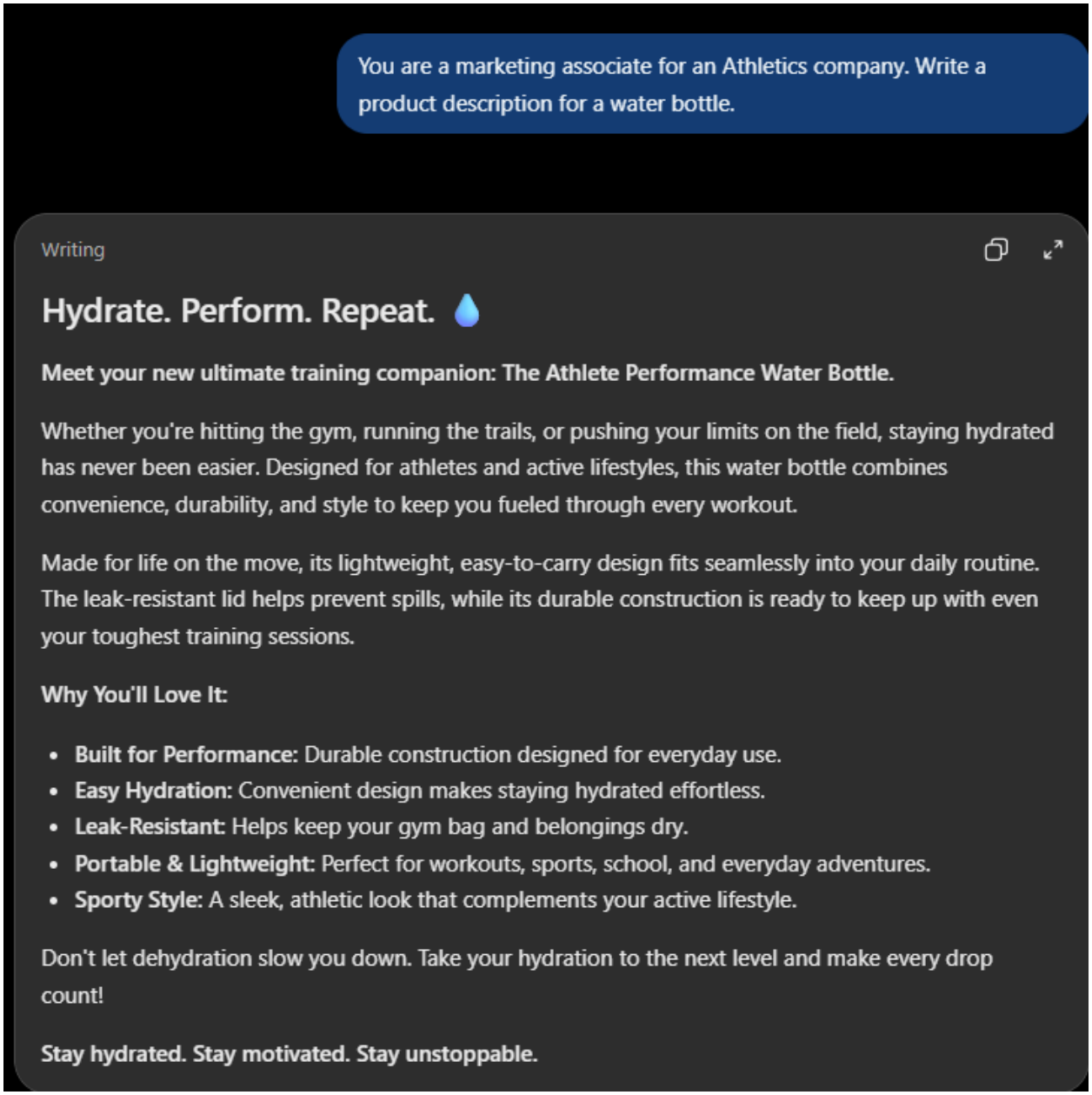
Part 1: Our initial experiment starts with the simple prompt below to ChatGPT.



Here we can see what sticks out right away - or what doesn't. The response is quite generic - this could describe *any* water bottle. While it does create a description for a water bottle, it describes traits that are applicable for *any* water bottle. However that can be attributed to our prompting - since we didn't provide any information or context, the model has very little to run off of. Basic questions like what're the dimensions? How much liquid can it hold? Is it plastic? Metal? Insulated? - are left unanswered. We haven't provided those details or what we're looking for, so naturally the output becomes a generic response as well.Iit doesn't seem to be written for a marketing team or someone looking to advertise their product, but more so for anyone looking for drafting inspiration or a general audience.

Part 2: Learning from our first response, we are going to try a mix of prompting techniques to make our end results more specific and informative.

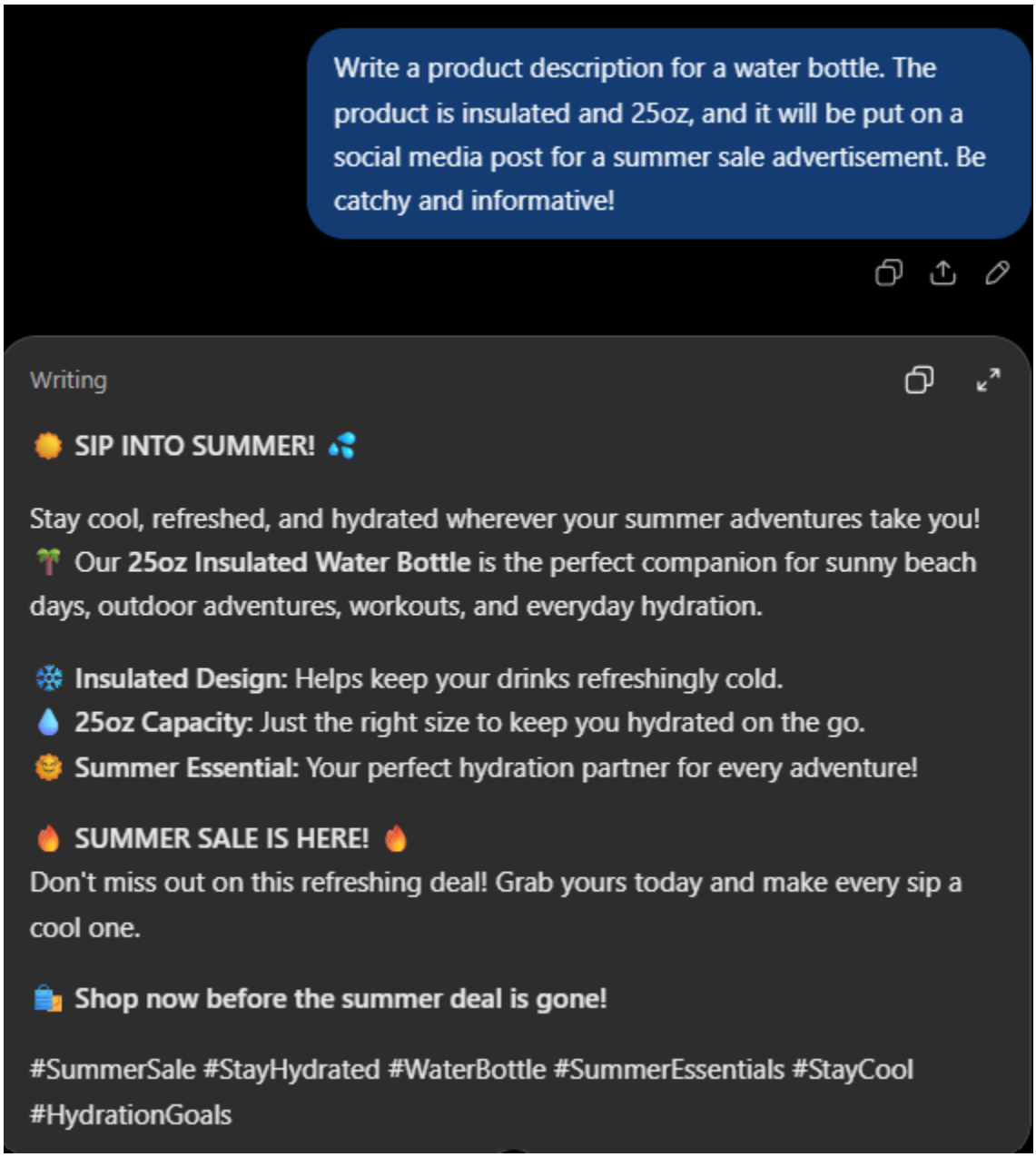
Water Bottle - Version 2 (Role Prompting)



The above technique was primarily **role prompting**, assigning the model as a ‘marketing associate’. While it doesn’t really become one, it definitely changes its output to at least accommodate the general idea of a marketing specialist - the sales pitch and advertising. In comparison to the first one, the formatting is much more readable and geared towards adding flair and liveliness with phrases such as “Why You’ll Love It” and other catchy slogans. While it’s

definitely more engaging, perhaps we can fine-tune this for more specific use cases with more emphasis on the next technique.

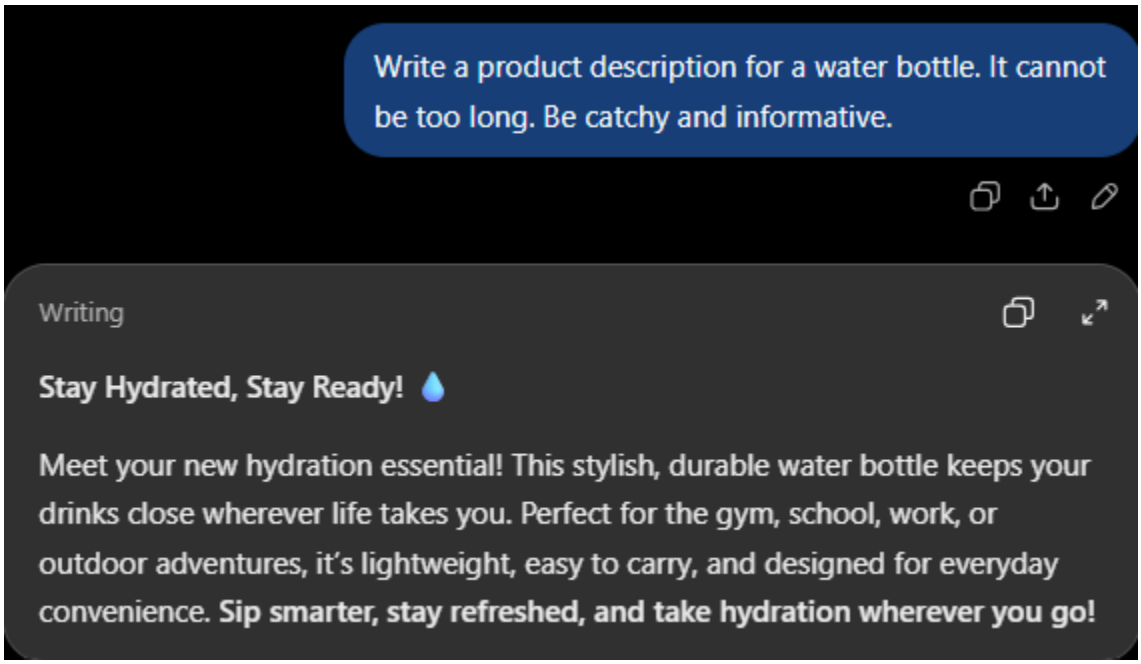
Water Bottle - Version 3 (Context)



As we can see, by putting an emphasis on **context**, the model was able to produce a much more creative and lively approach, adjusting to both our explicit and implicit context. It was not as lengthy as Version 2 because it had adjusted its approach to social media (where emojis are more accepted for advertising/descriptions), and included the specific information we requested it to incorporate - weather, sales, and the product information. This resulted in a great and lively

advertisement that wouldn't seem out of place for actual use - which shows how much control context can add to our outputs!

Water Bottle - Version 4 (Constraints)



Here we utilized **constraints** - and as we can see, it does follow the prompt and gives us what we explicitly put into it, but it does provide less control of the output. Telling it what we *don't* want - the default bland/general tone of the first response, can nudge the model in the right direction, but it's still a large range of freedom in comparison to providing context. Still very useful!

Water Bottle - Version 5 (Few-shot Prompting)



For this prompt we used **few-shot prompting**, where we gave the model the response from Version 2. As you can see - it delivered quite a similar structure but generalized it for a general audience. While useful, you can see an almost 1:1 copy in phrasing, sentence structure, format, and overall style. This is quite useful if you want very similar results in your output, but does provide less freedom by itself.

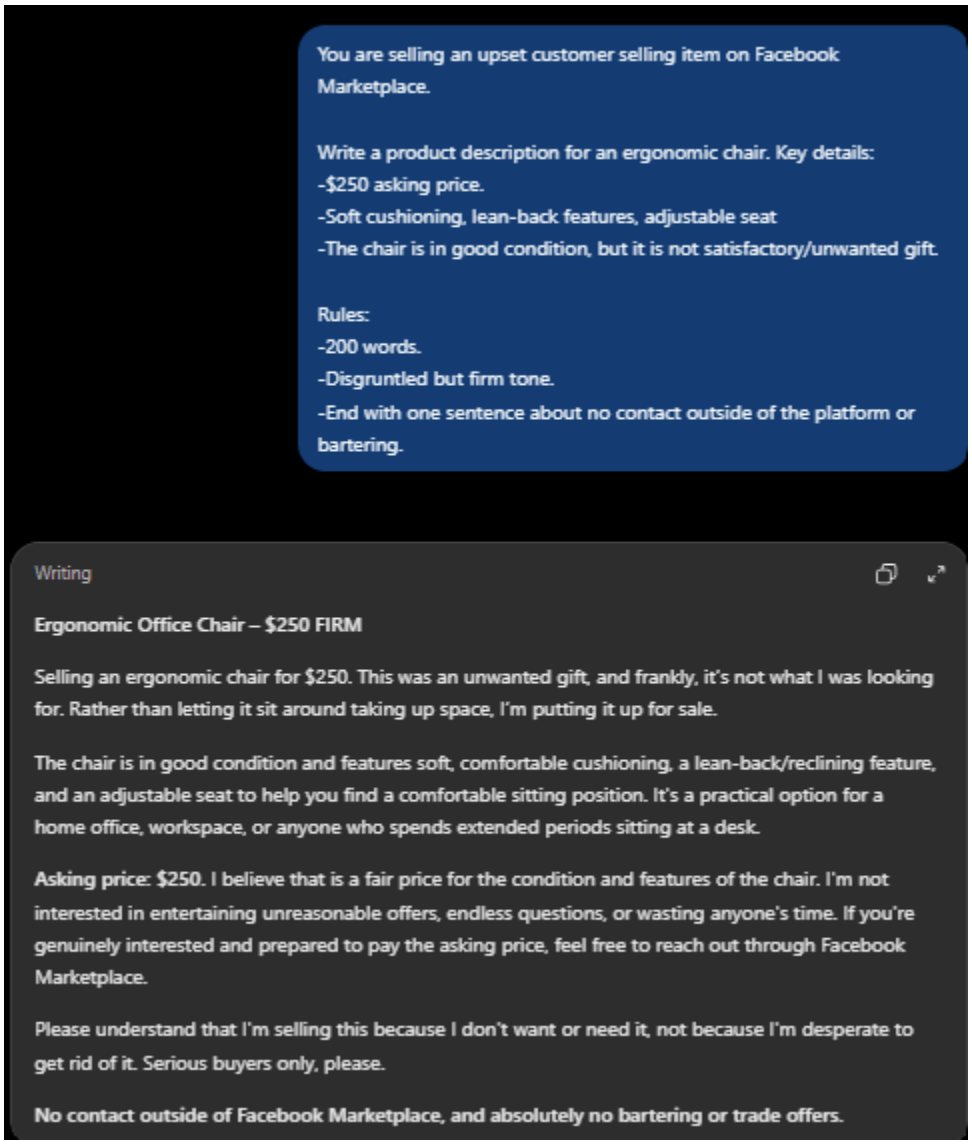
Part 3: New Products

Side note: I did toggle off memory and previous chat references, unsure if I had a particularly bland first response or something else happened. Strange.

Ergonomic Chair - Version 1

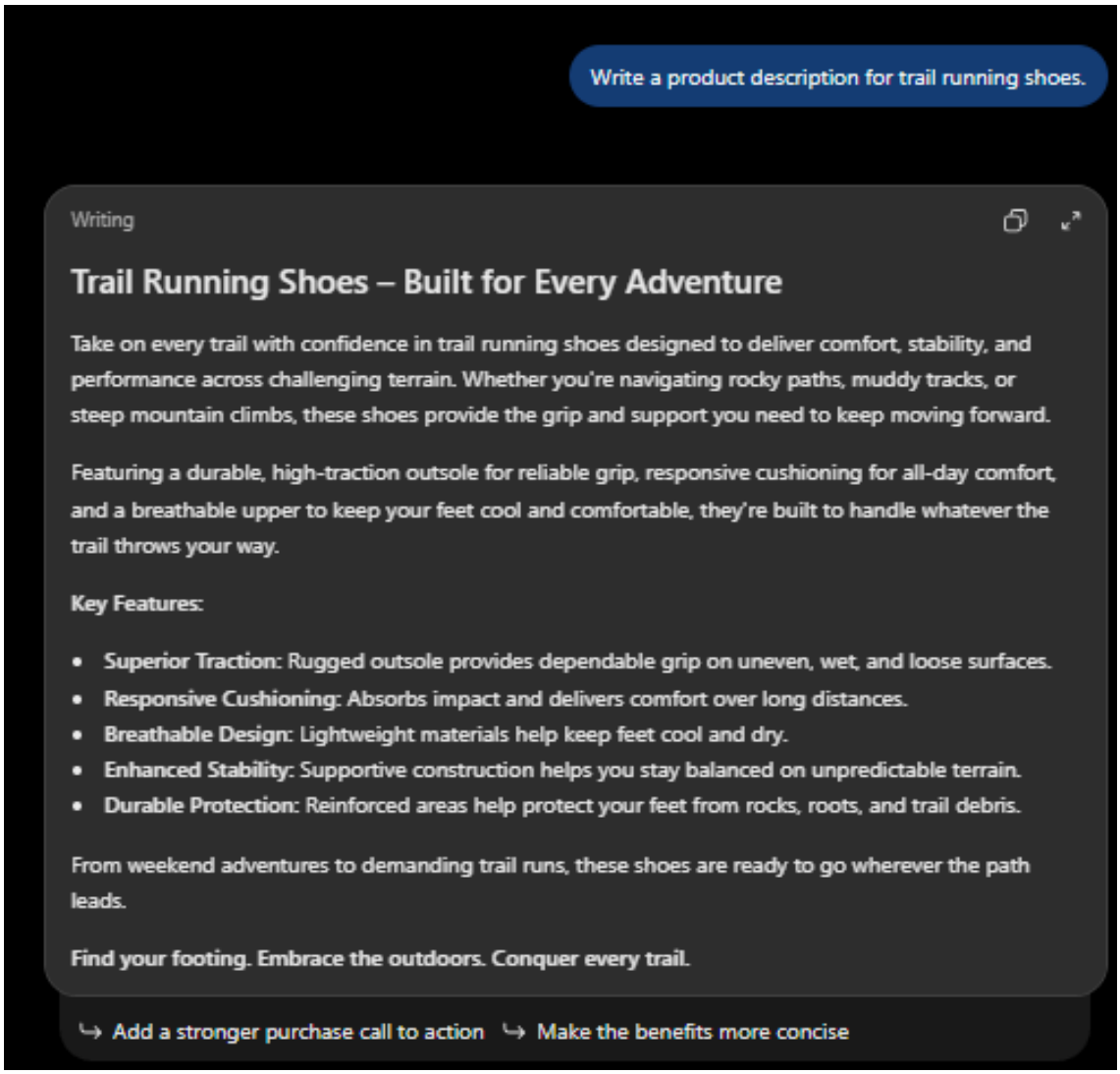


Ergonomic Chair - Version 2 (Role Prompting + Context + Constraints)

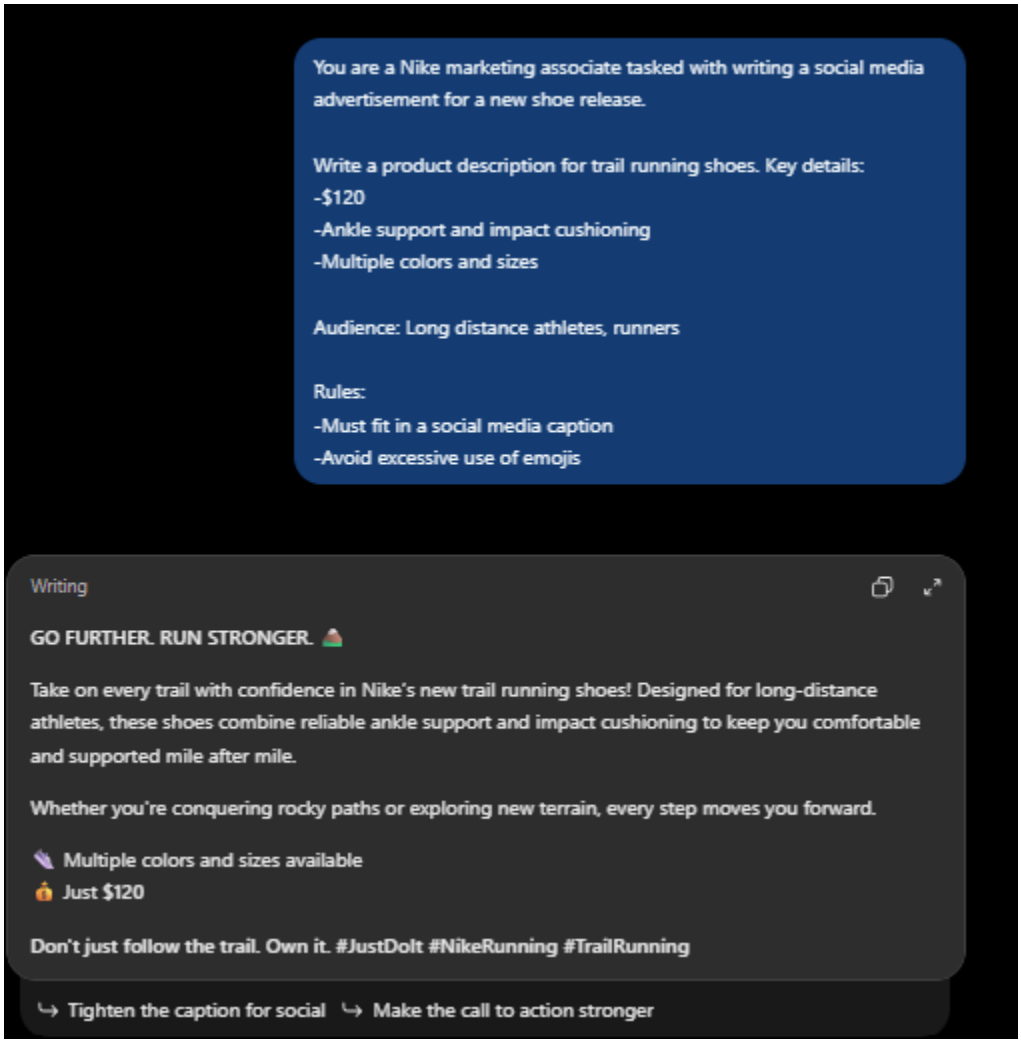


We can now see some drastic changes in a different direction. Given the role prompting - we have effectively changed the tone and delivery of the message to a more specific use case - emulating an upset customer. The context given also gave more information to the type of writing (formal/semi formal structure) and the restrictions gave it a specific tone for a marketplace ad on top of changing the formatting and delivery.

Trail Running Shoes - Version 1



Trail Running Shoes - Version 2 (Role prompting + Context + Constraints)



Here we see a key shift in the style - mainly the brevity and the adjustment to a social media platform. Given the role as a marketing associate, the result generated more of a sales-pitch ad. In addition the response is not as lengthy given the constraint, and mentioned the context of the price and design as well as incorporating a ‘Nike’-esque tone. Much more specific to our use-case compared to the generic response from before.

Analysis - Techniques
The techniques I primarily used and favored were role-prompting (given at the start with “You are xyz”), constraints (Rules are word limit, tone specifics, etc), and context (providing details around the scenario or product information). In isolation, I would rank the techniques from most to least impactful in response control - context, few-shot prompting, to role-prompting constraints, to output format.

Context I found the most useful in directing the model into what you would like it to output - adjusting the delivery from a generic response, to social media ad, to an online shop posting. The context provided not only incorporated many of the details requested, but was useful in implicit delivery - tone, structure, formatting, and other use cases.

Few-shot prompting gave the **most ‘literal’** control, but I noticed with a rigid sample - it would be **too** similar. From the water bottle Version 2 and Version 5, the structure and posting would be almost one to one - there wasn’t much variety. Of course, this can be useful if looking for the same responses, but may be limited if looking for creative use (drafting/inspiration).

Role prompting seemed to be a mixed bag, but of course can depend on the type of role and creativity of it. A disgruntled customer or Nike associate helped it change the tone and delivery of the outputs in creative ways.

Constraints were very helpful in explaining what you do and do not want - specific tones, restrictions and word limits, but in isolation they can be very limiting. There is still a lack of direction with only using this technique. Compare Water Bottles Version 4 and 1 - both are still somewhat bland, despite Version 4 adjusting to the rules.

Output format was quite powerful - the model follows these rules quite well, but if it’s only adjusting the formatting there is still a lack of direction. Useful for organizing.

In conjunction, all of these tools are very influential in dictating specific and useful responses.

Hallucinations/Model Overlooking details
Minor issues during my experiment - such as the Marketplace ad where the model would put too much emphasis on ‘disgruntled customer’. Providing more context on the product itself provided a much healthier balance on selling it without generating too much of an in-depth backstory. Hallucinations were most prevalent when the model had nothing to work with (solely the initial prompt) - so it made generic details about a non-existent product.

Overall
The takeaway from this is that it is important to direct your model into what you are looking for in most cases, and to refine your output in consideration of the limitations. Techniques are not end-all-be all, and a combination of them can help refine your inputs for more productive outputs.